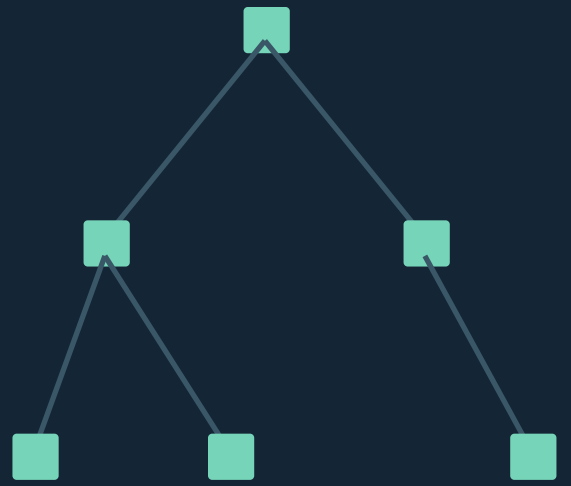
Testing LiTS with DeepSeek

How the allowed search steps affect correct answers and text use

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30 fixed mathematics questions

3 search limits: 1, 3, 10

9 completed repeated runs

Recorded experiment: 4 November 2026 | Presentation prepared for a 10 minute recording

What I reproduced

The authors' program proposes, checks and selects mathematical solutions.

01 **Read the question** > 02 **Propose solutions** > 03 **Check the options** > 04 **Select an answer**

PAPER / TABLE 6

37%

100 questions
Llama3-8B and the paper's scoring setup

THIS STUDY / LIMIT 10

85.6%

30 questions, 3 repeats
DeepSeek and API based scoring

Different model, scoring setup and sample size require a qualified comparison.

Sources: LiTS paper v2, Table 6 | assignment/experiment_plan.json | formal_results.json

One setting changes across three groups

The same questions and shared settings support the comparison.

SEARCH LIMIT

1

SEARCH LIMIT

3

SEARCH LIMIT

10

HELD FIXED

30 questions in source order
3 repeats with local seeds 11, 22, 33

10 search iterations / 3 alternatives
2 rollout steps / role specific decoding settings

ENFORCED IN EVERY GROUP

```
max_steps = 1, 3, 10
force_terminating_on_
depth_limit=True
```

Model: deepseek-flash
Thinking: disabled
Endpoint: api.deepseek.com

```
Source: 25e94d8
Data revision: d27dc61
```

Configuration excerpt, line wrapped for display | assignment/experiment_plan.json and assignment/study.py

Accuracy across all nine runs

Mean accuracy and sample SD from three repeats on the same 30 questions.

LIMIT 1

90.0% ± 0.0 pp

LIMIT 3

90.0% ± 3.3 pp

LIMIT 10

85.6% ± 5.1 pp

ACCURACY / SAME 30 QUESTIONS



ALL NINE CORRECT ANSWER COUNTS

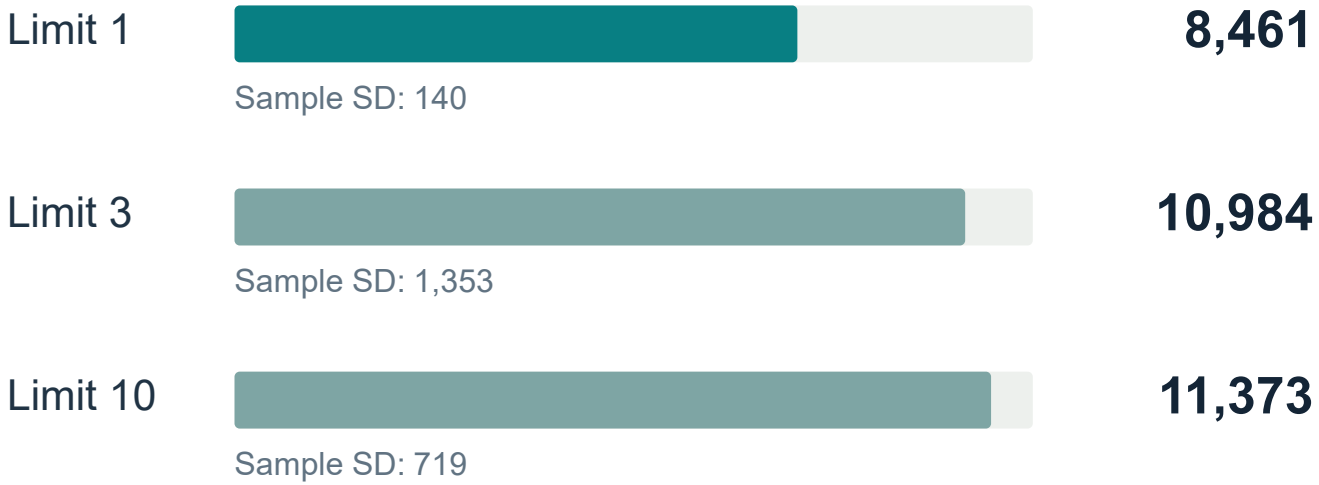
Limit	Run 1	Run 2	Run 3
1	27/30	27/30	27/30
3	28/30	27/30	26/30
10	24/30	26/30	27/30

Source: formal_trials.csv and formal_results.json | Error values describe variation between repeated runs.

Text use rises with the search limit

Counts include both search and the final answer selection.

TOKENS PER QUESTION / MEAN



26% lower

token use at limit 1 versus limit 10

RECORDED SECONDS / QUESTION

14.6 **16.9** **17.5**

Limits 1, 3, 10
Timing is descriptive: one retry ran alone; the first batch used 9 processes.

Source: formal_results.json | 4,425 formal model calls | Token counts represent consumption before price calculation.

One search step can include a full calculation

A single generated action may contain a complete calculation.

SAVED QUESTION 0 / LIMIT 1, RUN 1

How many positive whole numbers divide 196?

$$196 = 2^2 \times 7^2$$

$$(2 + 1) \times (2 + 1) = 9$$

Several mathematical operations
were completed inside one saved action.

CONFIGURED AND OBSERVED DEPTH

Limit	Mean*	Deepest
1	1.00	1
3	1.17	3
10	1.16	4

*Mean maximum depth per question.
The limit also controls when the final answer instruction is supplied.

Source: evidence/steps1_seed11/full_trees.jsonl, full_tree_0.json | Calculation reformatted from the saved action.

Repairs that made the experiment run

Compatibility changes were shared across all formal groups.

OBSERVED ISSUE	CHANGE	RECORDED OUTCOME
DeepSeek request options	Adjusted thinking and tool settings	9 selected formal runs completed with the shared API path.
Special character write failed	Used UTF-8 when saving text	The initial test and retained error record are documented.
Completion file replacement failed	Retained the attempt and repeated the entire test once	The other 8 completed runs were retained; the retry used identical source.

Nine selected runs have matching source hashes; the file replacement failure has an uncertain cause.

Source: assignment/DEBUG_DIARY.md, lits/lm/openai_chat.py, evidence/index.json and common_source_hashes.json

Saved records show how the experiment ran

Selected recorded fields are displayed here; the complete files remain in the repository.

COMPLETED RUN

Status
complete
Model
deepseek-flash
Search limit
1
Question count
30
Local seed
11

SAMPLE REQUEST

Request status
ok
Input tokens
455
Output tokens
121
Output limit
512
top_p / temperature
0.95 / 0.000001

SAVED OFFLINE CHECK

Status
passed
Python
3.12.12
Dataset rows
316
Formal rows
30
Model calls
0

Full records: settings, request use, evaluated answers, saved reasoning and excluded attempts

Sources: steps1_seed11/run_manifest.json and api_calls.jsonl, call 1 | Offline preflight: saved on 12 November 2026

Answer formatting affects the score

16 formatting related rejections were recorded across the 270 question evaluations.

ANSWER FORMAT / ORIGINAL CHECKER

Saved answer	Reference	Outcome
72°	72	Rejected
12 cm	12	Rejected
$78	78	Rejected

Primary scores retain the original checker results.

CONDITIONS THAT LIMIT CONCLUSIONS

30 distinct questions

An ordered subset with 5 pilot questions overlapping.

Online model variation

The model service and model based scoring can vary.

Different paper conditions

Model, scoring setup and sample size differ.

Source: formal_trials.csv, format_audit.json and experiment_plan.json | Diagnostic examples; primary scores remain unchanged.

Reproduce the test and use the evidence

The presentation contains the demonstration material for recording.

COMMANDS / IN AN ACTIVE PYTHON 3.12 ENVIRONMENT

01 Install the required packages

```
python -m pip install -r
    assignment/requirements-api.lock.txt
```

02 Prepare the fixed data

```
python -m assignment.prepare_data
```

03 Check the data and settings

```
python -m assignment.study preflight
```

04 Repeat the selected test

```
python -m assignment.study run --phase formal
    --max-steps 1 --seed 11
```

FINDINGS ON THIS SUBSET

90.0% accuracy at limit 1

26% lower token use versus limit 10

Measure search length, resource use and answer formatting.

github.com/programharu-creator/FTEC5660-Individual-Project-2-Reproducibility

Commands are line wrapped for display. Set your key in the ignored .env file; repeat limits 1, 3, 10 and seeds 11, 22, 33.

Full steps and offline evidence restoration: repository README | Recording link remains pending until recorded and shared.